MIL-P-52101D
11 March 1983
SUPERSEDING
MIL-P-52101C
28 November 1969

MILITARY SPECIFICATION

PLATES, BEARING AND COVER: FOR BRIDGE, FLOATING, ALUMINUM, HIGHWAY, DECK-BALK SUPERSTRUCTURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bearing and cover plates for the bridge, floating, aluminum, highway, deck-balk superstructure.

1.2 Classification. The plates shall be the following types, as specified (see 6.2):

Type I - Plate, Bearing, Balk-Connection Stiffener.
Type II - Plate, Cover, Long.
Type III - Plate, Cover, Short.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of specifications and standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings, Epoxy Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

D13213E4035	- Plate, Cover, Long.
D13213E4036	- Plate, Cover, Short.
D13213E4037	- Plate, Bearing, Balk-Connecting Stiffener.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:07Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers Inc., 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Standard Welding Code, Section 5, Qualification, and Section 9, Design of New Bridges.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 Northwest Seventh Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The plates shall be in accordance with the applicable drawings and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:07Z
Check the source to verify that this is the currentversion before use.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the plates. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Material deterioration and control. The plates shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify, the specific material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Stenciling. The nomenclature, national stock number, weight, and cubage shall be stenciled on each plate. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.5 Type I plate, bearing, balk-connecting stiffener. The type I plate shall be as shown on Drawing D13213E4037.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:07Z
Check the source to verify that this is the current version before use.

3.6 Type II plate, cover, long. The type II plate shall be as shown on Drawing D13213E4035.

3.7 Type III plate, cover, short. The type III plate shall be as shown on Drawing D13213E4036.

3.8 Workmanship. The fabricated plates shall be free from sharp edges, slivers, or burrs.

3.8.1 Steel fabrication. Steel used in the fabrication of the plates shall be free from laminations, kinks, and sharp bends. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, with a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. The straightening of material shall be done by methods that will not cause injury to the metal. Precautions shall be taken to avoid overheating,and heated metal shall be allowed to cool slowly.

3.8.2 Aluminum fabrication. Before any work or layout is done, aluminum shall be straightened by methods that will not result in injury to the metal. Care shall be used in handling aluminum during fabrication to avoid excessive scratching or damage to edges. Finished members shall be free from laminations, kinks, bends, scratches, or winds. Corners shall be square and true. All bends shall be made to a radius which will not cause cracking in the extreme outer fibers. Bends shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines. Aluminum 3/8 inch thick and over shall be cut by routing or sawing. Aluminum under 3/8 inch thick may be cut by shearing. Aluminum shall not be cut by methods using heat.

3.8.3 Riveting. Aluminum rivets shall be driven cold. Heads of driven rivets shall be uniform in shape, neatly formed, and concentric with rivet shank axis, and shall have full bearing on the surfaces of the jointed members. Calking or redriving of rivet heads shall not be permitted. Rivets with a head deficient in size, not uniform in shape, offcenter, loose, or misshapen in any way shall be considered defective. Defective rivets shall be removed without injury to the joined metal. Where practicable, rivets shall be driven by direct-action rivet machines. In lieu of direct-action rivet machines, pneumatic rivet hammers may be used. Excessive air-driving pressures or over-driving of rivets resulting in damage to the joined members shall not be permitted (see 6.5.2).

3.8.4 Welders and steel welding.

3.8.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
AWS D1.1, Structural Welding Code, Section 5, Qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

3.8.4.2 Steel welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded, electric-arc method and shall conform to the provisions of the current American Welding Society publication D1.1, Section 9 and as specified herein. Work shall be positioned for flat welding whenever practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean. The finished item shall be straight and within toleranced dimensions. Straightening and stress relieving shall be permitted to relieve distortion, provided the straightening process does not damage the end item.

3.8.5 Interchangeability. Like units shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

3.8.6 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings. Tolerances for positioning of holes and parts shall be referenced from one end or side of the member and shall not be cumulative.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Examination. The first article plate shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The plates shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the QAP requirements shown on the drawings.
102. Material not as specified.
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatments.
106. Workmanship not as specified.
107. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Stenciling not as specified.
203. Dimensions other than those identified on the QAP not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Type II and type III plates not bundled in a manner to prevent damage to the lugs as specified for level A.
109.	Strapping not as specified for level A.
110.	Marking illegible, incomplete, incorrect, or missing for level A or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Type I plates shall be shipped loose or bundled in quantities most convenient for handling during shipment. Type II and III plates of like description shall be bundled, in a manner to prevent damage to the lugs, in quantities most convenient for handling during shipment. Bundled plates shall be secured with flat steel strapping conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.1.2 Commercial. The plates shall be prepared for shipment loose or bundled as specified in 5.1.1. Bundled plates shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in bundles complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Loose and bundled plates shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bearing plate is intended for use on the abutment of the support of the ramp of bridges using deck-balk superstructure. The short and long cover plates are used to cover the gap at the junction of the ramp and the deck of the bridge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of plate required (see 1.2).
c. When a first article is required for inspection and approval and the number of units required (see 3.2, 4.3, 6.3).
d. Degree of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be examined and should be a first produced item. The first article should consist of one or more units. The contracting officer should include specific instructions for examination and approval of the documents first article.

6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.5.2 Rivet head. Rivets have a manufactured head or a shop head. The diameter of a correctly formed shop head should be 1-1/2 diameters of the rivet shank and the height one-half the diameter. A scaled rivet gage should be used to check the upset rivet head to see that it conforms to requirements. Manufactured heads can be visually checked for deformities such as tipping or looseness.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 5420-0142

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:07Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-P-52101D Plates, Bearing and Cover: For Bridge, Floating, Aluminum, Highway, Deck-Balk Superstructure

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:07Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY MOBILITY EQUIPMENT RESEARCH
AND DEVELOPMENT COMMAND
ATTN: DRDME-DS
FORT BELVOIR, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:07Z
Check the source to verify that this is the current version before use.